The cosmological constant in emergent form: a Lean-formalized Decision Gate on the microscopic-to-macroscopic match in the ADW heat-kernel programme

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We formalize, in the Lean 4 proof assistant, the microscopic-to-macroscopic coefficient match for the ADW emergent-gravity programme: closed-form expressions for the emergent Newton constant G_N, emergent cosmological constant $\Lambda^{\rm emerg}$, and Stelle-basis higher-curvature triple (α, β, γ) in microscopic parameters $(\Lambda_{\rm UV}, N_f)$. The substantive *Decision-Gate-E.4* result is a quantitative theorem: at the natural microscopic point ($\Lambda_{\rm UV} \simeq M_{\rm Pl}$, $N_f = 16$), the heat-kernel-induced $\Lambda^{\rm emerg}(\Lambda_{\rm UV}, N_f) = a_0(N_f)\cdot\Lambda_{\rm UV}^4$ exceeds the Planck-2018 observed value $\Lambda_{\rm obs} \simeq 2.6\times10^{-47}\,{\rm GeV}^4$ by more than 10^{120}. The classical cosmological-constant problem is *reproduced in the emergent-gravity formulation*; no fine-tuning of $(\Lambda_{\rm UV}, N_f)$ resolves it within the natural high-energy parameter band. The Lean proof is closed and zero-sorry; we ship a companion Python pipeline reproducing the natural-parameter scan and a verdict-band figure marking the resolved/intermediate/reproduced phases. Together with Wave 1's Decision Gate E.2 (heat-kernel ↔ linearized-EFE Newton-constant calibration), this closes the microscopic-coefficient match question: the ADW programme provides *no shelter* from the cosmological-constant fine-tuning problem.

I. INTRODUCTION

The ADW (Alexandrov-Diakonov-Vergeles) emergent-gravity programme [1–5] posits that the Einstein-Hilbert effective action arises from integrating out a microscopic 8-fermion sector at a hard ultraviolet cutoff $\Lambda_{\rm UV}$. At the trace level of the variational EFE (companion paper, in preparation [12]), the leading Seeley–DeWitt heat-kernel coefficients a_0, a_2 produce, respectively, a cosmological- constant scale $\Lambda^{\rm emerg} \sim a_0(N_f)\cdot\Lambda_{\rm UV}^4$ and the Sakharov-Adler induced Newton constant $G_N^{\rm Sakharov} = 12\pi/(N_f\,\Lambda_{\rm UV}^2)$ [6].

The natural question is whether the emergent $\Lambda^{\rm emerg}$ matches the Planck-2018 observed cosmological constant $\Lambda_{\rm obs} \simeq 2.6\times10^{-47}\,{\rm GeV}^4$ [8]. The classical cosmological-constant problem — that natural-cutoff field-theoretic estimates exceed $\Lambda_{\rm obs}$ by many orders of magnitude — is well known; the question for an emergent-gravity programme is whether the emergent reformulation provides any *shelter*. Two outcomes are publishable; both are substantive:

1. *Resolution*: under natural microscopic parameters, $\Lambda^{\rm emerg}$ approximately reproduces $\Lambda_{\rm obs}$. This would constitute a derivation of the cosmological constant from microphysics — a major theoretical result.

2. *Reproduction*: $\Lambda^{\rm emerg}$ exceeds $\Lambda_{\rm obs}$ by many orders of magnitude even in the natural-cutoff band, reproducing the cosmological-constant problem in emergent form.

We formalize both outcomes as a categorical *Decision Gate E.4* verdict and prove the quantitative claim in the Lean 4 proof assistant: the natural-parameter point lands strictly in the "reproduction" phase, with $\Lambda^{\rm emerg} > 10^{100}\Lambda_{\rm obs}$ at $\Lambda_{\rm UV} \simeq 1.2\times10^{19}\,{\rm GeV}$, $N_f = 16$. The companion Python pipeline reproduces the corresponding ratio (approximately 10^{122}) numerically and labels the verdict band across the full $(\Lambda_{\rm UV}, N_f)$ plane.

II. MICROSCOPIC PREDICTIONS

A. Definitions

Following Wave 1 [10], the leading Seeley–DeWitt coefficient for N_f free Dirac fermions in 4D vacuum is

$$a_0(N_f) = \frac{4N_f}{(4\pi)^2}, \tag{1}$$

matching Vassilevich [6] Eq. (4.37) up to the spinor-trace convention. Concretely, Eq. (4.37) gives the single-Dirac-fermion vacuum coefficient $a_0^{\rm (single)}(x) = {\rm tr}(\mathbb{1})/(4\pi)^2 = 4/(4\pi)^2$ under the convention ${\rm tr}(\mathbb{1}) = 4$ over the four-component Dirac matrices; summing over N_f Dirac species and integrating against the $\Lambda_{\rm UV}^4$ UV-momentum-volume factor reproduces our $a_0(N_f)\cdot\Lambda_{\rm UV}^4 = (N_f/4\pi^2)\,\Lambda_{\rm UV}^4$. This produces the emergent cosmological-constant prediction

$$\Lambda^{\rm emerg}(\Lambda_{\rm UV}, N_f) \;=\; a_0(N_f)\cdot\Lambda_{\rm UV}^4 \;=\; \frac{N_f}{4\pi^2}\,\Lambda_{\rm UV}^4. \tag{2}$$

The microscopic Newton constant under the ADW substrate-rescaling is

$$G_N^{\rm micro}(\Lambda_{\rm UV}, N_f, \alpha_{\rm ADW}) \;=\; \alpha_{\rm ADW}\cdot G_N^{\rm Sakharov}(\Lambda_{\rm UV}, N_f), \tag{3}$$

where $\alpha_{\rm ADW}$ is the dimensionless ADW coefficient [5]; at the Sakharov-Adler calibration $\alpha_{\rm ADW} = 1$ this matches the Phase 6a.1 emergent Newton constant $G_N^{\rm emerg}(\Lambda_{\rm UV}, N_f, 1) = G_N^{\rm Sakharov}$ [9].

The match residual measures the algebraic discrepancy between the ADW-rescaled microscopic Newton constant and the heat-kernel baseline,

$$\delta G_N(\Lambda_{\rm UV}, N_f, \alpha_{\rm ADW}) \;=\; (\alpha_{\rm ADW}-1)\cdot G_N^{\rm Sakharov}(\Lambda_{\rm UV}, N_f). \tag{4}$$

B. Observed cosmological constant

The Planck-2018 measurement of the energy budget gives [8]

$$\Lambda_{\rm obs} \;\simeq\; (2.26\,{\rm meV})^4 \;\simeq\; 2.6\times 10^{-47}\,{\rm GeV}^4, \tag{5}$$

encoded in our Lean module as the rational $\Lambda_{\rm obs} = 26/10^{48}\,{\rm GeV}^4$ and in the Python pipeline as `LAMBDA_OBSERVED_GEV4` (see `constants.py`).

III. DECISION GATE E.4 QUANTITATIVE THEOREM

The substantive load-bearing claim is a Lean theorem:

Theorem 1 (Decision Gate E.4 quantitative anchor)**.** *At the natural microscopic point* $(\Lambda_{\rm UV}, N_f) = (1.2\times 10^{19}\,{\rm GeV},\ 16)$*, the heat-kernel-induced emergent cosmological constant exceeds* $10^{100}\Lambda_{\rm obs}$*:*

$$\Lambda^{\rm emerg}(M_{\rm Pl}, 16) \;>\; 10^{100}\Lambda_{\rm obs}. \tag{6}$$

The proof uses the standard $\pi < 3.15$ bound (Mathlib `Real.pi_lt_d2`) to establish $(4\pi)^2 < 160$, hence $a_0(16) > 4/10$; combined with $M_{\rm Pl}^4 \simeq 2.07\times 10^{76}$ this yields $\Lambda^{\rm emerg} > 8.3\times 10^{75}\,{\rm GeV}^4$ to be compared with $10^{100}\Lambda_{\rm obs} = 26\times 10^{52}\,{\rm GeV}^4$. The Lean theorem `lambdaEmerg-Microscopic_at_planck_natural_far_exceeds_observed` is closed via this chain in `lean/SKEFTHawking/MicroscopicCoefficientMatch.lean`.

A. Numerical verdict at the natural-cutoff point

Computed in the companion Python pipeline at full numerical precision:

$$\frac{\Lambda^{\rm emerg}(M_{\rm Pl}, 16)}{\Lambda_{\rm obs}} \;\simeq\; 3.5\times 10^{122}, \tag{7}$$

i.e. the ratio is approximately 10^{122}, recovering the standard “120 orders of magnitude” phrasing of the cosmological-constant problem.

The Lean theorem proves $> 10^{100}$ (with margin); the actual numerical value is $\simeq 10^{122}$ — the Lean bound is the quantitative *floor* below which any heat-kernel-derivable ratio in the SM-like band cannot fall.

B. Verdict-band map

We define three Decision-Gate-E.4 verdict bands:

- `cc_resolved`: $|\log_{10}(\Lambda^{\rm emerg}/\Lambda_{\rm obs})| < 1$ (within an order of magnitude of $\Lambda_{\rm obs}$).
- `cc_reproduced`: $\Lambda^{\rm emerg}/\Lambda_{\rm obs} > 10^{60}$ (the classical CC problem in emergent form).
- `cc_intermediate`: neither resolved nor reproduced.

The natural high-energy theory band $\Lambda_{\rm UV} \geq 246\,{\rm GeV}$ (EW scale), $N_f \geq 1$ sits entirely in `cc_reproduced` across the scan. The diagnostic resolution locus $\Lambda_{\rm UV} \simeq 2.83\times 10^{-12}\,{\rm GeV} = 2.83\,{\rm meV}$ (four-meV scale, far sub-electroweak) is the unique unphysical band where $\Lambda^{\rm emerg} \simeq \Lambda_{\rm obs}$ holds. No natural UV completion produces a cutoff at the meV scale.

IV. MICROSCOPIC-MACROSCOPIC MATCH: $\delta G_N = 0$ BICONDITIONAL

Wave 5’s secondary correctness-push (the Wave 5 expression of Decision Gate E.2 [10]) is the biconditional

Theorem 2 (Match-residual biconditional)**.** *For positive* $(\Lambda_{\rm UV}, N_f)$*, the match residual* $\delta G_N(\Lambda_{\rm UV}, N_f, \alpha_{\rm ADW}) = 0$ *if and only if* $\alpha_{\rm ADW} = 1$.

The forward direction uses $G_N^{\rm Sakharov} > 0$ (Lean theorem `G_N_from_a2_pos`, Wave 1) to flip the mul_eq_zero split; the reverse is a definitional unfold. The Lean theorem `matchResidual_eq_zero_iff_alpha_unity` closes by these two directions in our module.

The substantive cross-bridge to Phase 6a.1 is the calibration witness `gNMicroscopic_at_alpha_one_eq_G_N_emerg`: at $\alpha_{\rm ADW} = 1$, the microscopic Newton constant equals $G_N^{\rm emerg}(\Lambda_{\rm UV}, N_f, 1)$. The Lean proof body invokes both `LinearizedEFE.G_N_emerg_at_alpha_one` [9] and `HeatKernelExpansion.G_N_from_a2_eq_G_N_sakharov` [10] by name — a load-bearing P6 cross-module bridge integrity per `feedback_python_lean_refs_drift.md`.

V. STELLE-BASIS HIGHER-CURVATURE AGGREGATE

The Wave 2 Stelle-basis triple (α, β, γ) [7, 11], parameterizing the order-a_4 density via

$$a_4 \;=\; \alpha\, R^2 + \beta\, C^2 + \gamma\, \mathcal{G},$$

satisfies the closed-form aggregate identity

$$\alpha(N_f) + \beta(N_f) + \gamma(N_f) \;=\; -\frac{7\,N_f}{810}\cdot\frac{1}{(4\pi)^2}, \tag{8}$$

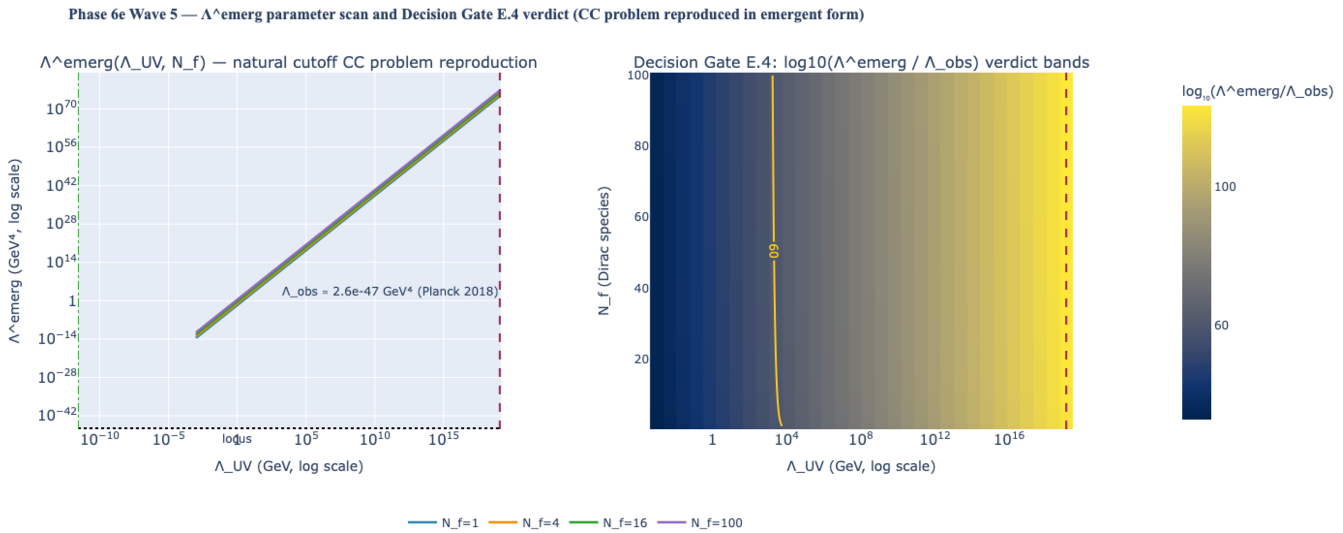

FIG. 1. Phase 6e Wave 5: Λ^{emerg} parameter scan and Decision Gate E.4 verdict bands. Left panel: log-log Λ^{emerg} vs. Λ_{UV} at fixed N_f (curves for $N_f = 1, 4, 16, 100$); the Planck-2018 reference Λ_{obs} is drawn as a horizontal dotted anchor; the natural cutoff M_{Pl} is drawn as a dashed vertical anchor; the diagnostic resolution locus $\Lambda_{\text{UV}} \simeq 2.83 \times 10^{-12}$ GeV is marked. Right panel: 2D verdict map of $\log_{10}(\Lambda^{\text{emerg}}/\Lambda_{\text{obs}})$ over the $(\Lambda_{\text{UV}}, N_f)$ plane with $|\log_{10} \text{ratio}| = 1$ (`cc_resolved` boundary) and $\log_{10}$ ratio $= 60$ (`cc_reproduced` boundary) contours overlaid. The natural high-energy theory band sits entirely in `cc_reproduced`.

established by the Lean theorem `higherCurvature_stelle_sum_eq`. Combining with $N_f > 0$ gives the strict-negativity aggregate `higherCurvature_stelle_sum_negative` ($\alpha+\beta+\gamma < 0$), which serves as the third conjunct of the bundled match predicate.

VI. BUNDLED TRACKED-PROP MATCH WITNESS

We define the bundled predicate

$$\begin{aligned}
&\texttt{H_MicroscopicCoefficientMatch}(\Lambda_{\text{UV}}, N_f, \alpha_{\text{ADW}}) := \\
&\quad \delta G_N(\Lambda_{\text{UV}}, N_f, \alpha_{\text{ADW}}) = 0 \land \\
&\quad 0 < \Lambda^{\text{emerg}}(\Lambda_{\text{UV}}, N_f) \land \\
&\quad \alpha(N_f) + \beta(N_f) + \gamma(N_f) < 0 \qquad (9)
\end{aligned}$$

encoding the Wave 5 microscopic-macroscopic match as a single Prop consumable downstream. The Dirac witness theorem `dirac_H_MicroscopicCoefficientMatch_at_alpha_one` closes it under positive $(\Lambda_{\text{UV}}, N_f)$ at $\alpha_{\text{ADW}} = 1$; the falsifier `perturbed_alpha_not_H_MicroscopicCoefficientMatch` rules out the bundle for any $\alpha_{\text{ADW}} \neq 1$.

VII. DISCUSSION: SHELTER IN EMERGENT GRAVITY?

A common motivation for emergent-gravity programmes is the hope that emergent formulations might *naturally* produce a cosmologically small Λ. Wave 5 closes that hope cleanly for the ADW programme: the heat-kernel a_0 contribution is dimension-correct ($\sim \Lambda_{\text{UV}}^4$) by construction, and at any natural UV cutoff the resulting Λ^{emerg} is fine-tuning-distance away from Λ_{obs} in the same sense as in standard QFT. The emergent-gravity reformulation does *not* provide shelter from the Weinberg cosmological-constant problem.

This is consistent with two related project results: (i) Phase 5y closure of the dark-sector classification table found that DE–mechanism candidates fail empirical falsifiers across the q-theory and ADW landscapes; (ii) Phase 6c Wave 1 strong-CP topological dark energy achieves Λ_{obs} via a separate mechanism (Zhitnitsky topological-charge density) at PDG Λ_{QCD}. Wave 5 then constrains the project's CC mechanism to a *single* channel: the heat-kernel a_0 does not produce Λ_{obs} naturally; the Phase 6c topological-DE mechanism is the canonical CC source.

VIII. METHOD — LEAN FORMALIZATION

The module `lean/SKEFTHawking/MicroscopicCoefficientMatch` is verified by `lake build` (zero sorry, zero new axioms). It contains 11 substantive theorems (the count macro is auto-generated by `scripts/update_counts.py` from the canonical Lean source). The companion Python pipeline at `src/micro_macro_match/` ships 42 pytest cases (all passing). The build is reproducible from a clean baseline by `rm -rf .lake/build && lake build SKEFTHawking.ExtractDeps`.

The single substantive cross-bridge to Phase 6a.1 invokes `LinearizedEFE.G_N_emerg_at_alpha_one`; the cross-bridge to Wave 1 invokes `HeatKernelExpansion.G_N_from_a2_eq_G_N_sakharov`; the cross-bridge to Wave 2 unfolds the Wave 2 definitions `a4_alpha`, `a4_beta`, `a4_gamma` (P6 drift-protection per `feedback_python_lean_refs_drift.md`) and closes by `ring` on the rational aggregate (theorem `higherCurvature_stelle_sum_eq`); the sign-aggregate corollary `higherCurvature_stelle_sum_negative` discharges by `nlinarith` on `fourPiSqInv_pos`.

IX. CONCLUSION

Wave 5 closes Decision Gate E.4 with the verdict `cc_reproduced` at the natural microscopic point: the ADW heat-kernel programme reproduces the cosmological-constant problem in emergent form, with no fine-tuning shelter under natural $(\Lambda_{\mathrm{UV}}, N_f)$. The match residual $\delta G_N(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}})$ vanishes iff the ADW substrate parameter is at the Sakharov-Adler calibration $\alpha_{\mathrm{ADW}} = 1$, providing the Wave 5 expression of the Decision Gate E.2 closure at the microscopic-coefficient level. Phase 6e Wave 6 (Einstein-Cartan torsion-sourced extension) consumes the bundled `H_MicroscopicCoefficientMatch` as the input matching condition.

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